HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained high-performance C++ applications for vehicle edge computing systems, focusing on data processing and real-time performance.
- Authored clean, efficient, and maintainable C++17 code, adhering to modern software development practices and design patterns.
- Implemented comprehensive unit and integration tests using C++ with **Google Test**, achieving over **90% code coverage** for critical application modules.
- Collaborated with **cross-functional teams** including QA and DevOps to integrate vehicle application features into the broader software-defined vehicle platform.
- Participated in **code reviews**, providing constructive feedback to peers and ensuring adherence to coding standards and best practices.
- Debugged and resolved complex software issues within a **Linux environment**, performing root cause analysis and implementing robust fixes.
- Wrote **Python scripts** to automate build processes and tooling, streamlining development workflows for the vehicle applications team.

SKILLS

Languages – C++, Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, C++11, C++17, Google Test, scikit-learn, pandas, NumPy

Tools & Technologies – AWS, Docker, Git, Jenkins, Jira, Confluence, CMake, Linux

Databases – PostgreSQL, MySQL, Oracle

Operating Systems – Linux

Methodologies – Agile, Scrum, CI/CD

PROJECTS

Real-Time Vehicle Data Processing Engine

- Designed and built a C++ application for processing high-velocity sensor data within an automotive edge computing context, handling over **5,000 events per second**.
- Implemented multi-threading and concurrency primitives to ensure efficient data ingestion and parallel processing, reducing data latency by **30%**.
- Developed a **CMake** build system for the C++ application, ensuring cross-platform compatibility and streamlined compilation across development and target environments.

Automated System Health Monitoring Tool

- Created a Python-based tool that monitors the health of embedded vehicle systems, parsing logs from **Linux** environments and identifying potential issues.
- Integrated the tool with **Jenkins** to run automated checks as part of the CI pipeline, catching configuration drift and critical errors before deployment.
- Generated daily summary reports detailing system status and anomalies, which were used by the operations team to proactively address issues.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)